

Module: ADS1 Algorithms and Data Structure 1
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Type: Tutorial

TD 03: Loops

Exercise 1

Given the following algorithm:

Algorithm 1:

```
Algorithm Countdown;  
Var: I: integer;  
begin  
  I  $\leftarrow$  10;  
  while I  $\geq$  1 do  
    Display(I);  
    I  $\leftarrow$  I - 1;  
  end  
  Display("Lift off!");  
end
```

- What is the main objective of this algorithm?
- How many times will the loop execute?
- What will happen if the initial value is changed to 5?
- Rewrite the same logic using a Repeat...Until structure.

Exercise 2

Given the following algorithm:

Algorithm 2:

```
Algorithm AverageScore;  
Var: Score, Total, I, N: real;  
begin  
  Total  $\leftarrow$  0;  
  for I  $\leftarrow$  1 to 5 do  
    Read(Score);  
    Total  $\leftarrow$  Total + Score;  
  end  
  Display("Average = ", Total / 5);  
end
```

- What is the purpose of this algorithm?
- What kind of loop is used?
- Modify it to read the number of students dynamically (ask the user how many).

Exercise 3

Write an algorithm that reads a series of grades (0-20) and counts how many students have:

- Passed (grade $\geq$ 10)
- Failed (grade $<$ 10)
- Add one more output: the success rate in %.

The loop should stop when the user enters a negative number.

Exercise 4

Write an algorithm that reads 10 integers and determines:

- the maximum value,
- the minimum value,
- and the average of all entered numbers.

Exercise 5

Write an algorithm that asks the user to enter a number N, then calculates the **sum of the first N odd numbers** ($1 + 3 + 5 + \dots$).

Example: If $N = 4 \rightarrow \text{sum} = 1 + 3 + 5 + 7 = 16$

Exercise 6

Write an algorithm that simulates a **simple password system with 3 attempts**:

- Ask the user to enter a password.
- If it is correct $\rightarrow$ display "Access Granted".
- If not $\rightarrow$ repeat until the third wrong attempt, then display "Access Denied".

Exercise 7

Write an algorithm that reads an integer **N** and computes its **factorial**.

Example:

$$N = 5 \Rightarrow 5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$$

Exercise 8

Write an algorithm that reads an integer **N** and calculates the **sum of its digits**.

Example: $N = 753 \rightarrow \text{Sum} = 7 + 5 + 3 = 15$

Exercise 9

Write an algorithm that reads an integer **N** and displays its digits in **reverse order**.

Example: $N = 1234 \rightarrow \text{Output} = 4321$

Exercise 10

A number is called an **Armstrong number** if the sum of the cubes of its digits equals the number itself. **Example:** $153 = 1^3 + 5^3 + 3^3$

Write an algorithm that checks whether a number **N** is an Armstrong number or not.